Jiacheng Lai

JiachengLai@outlook.com | Homepage | GitHub

Education

School of Mathematics | Changsha University Sept. 2023 - Present
B.S. in Mathematics and Applied Mathematics

Employment

Shenzhen University of Advanced Technology Jul. 2026 - Present
Research Intern | Advisor: Hui Peng

Bioinformatics Complex Network Big Data Research Institute | Changsha University Oct. 2023 - Jul. 2026
Undergraduate Research Assistant | Advisor: Lei Wang

Research Interest

Biological systems are rarely observed from a single complete perspective; instead, they are reflected through diverse and often imperfect data modalities, including molecular structures, protein sequences, microbial genomic signals, biological networks, drug-response profiles, and multi-omics measurements. Motivated by this challenge, my research focuses on reliable and calibrated machine learning for computational biology and drug discovery. My broader interest lies in applying mathematical, statistical, and machine learning principles to fundamental problems in bioinformatics, including heterogeneous data integration, uncertainty quantification, low-data prediction, and reliable inference from noisy and incomplete biomedical data. My previous work includes heterogeneous graph-based microbe-drug association prediction, feature-independent cold-start drug-microbe association learning, and multimodal drug-Mtb affinity prediction for anti-tuberculosis drug repurposing.

Publications and Manuscripts

(* stands for equal contribution.)

[1] *(Journal of Molecular Biology 2025)* Jiacheng Lai, Zhen Zhang, Bin Zeng, and Lei Wang. "GTFKAN: A novel microbe-drug association prediction model based on graph transformer and Fourier Kolmogorov-Arnold networks." **DOI** | **Code**

TL;DR | Developed GTFKAN, a hybrid model combining Graph Transformer Networks and Fourier Kolmogorov-Arnold Networks to predict microbe-drug associations by integrating heterogeneous network structures and high-order latent features..

[2] *(Under Review)* Jiacheng Lai, Zhen Zhang, Hao Yuan, and Lei Wang. "BMFN: Multimodal feature fusion network for drug-Mtb affinity prediction and anti-tuberculosis drug repurposing." **Code**

TL;DR | Developed BMFN, a multimodal fusion network for drug-Mycobacterium tuberculosis target affinity prediction, integrating sequence, heterogeneous graph, and molecular fingerprint features to support anti-tuberculosis drug repurposing and candidate prioritization.

[3] *(Submitted to IEEE BIBM 2026)* Jiacheng Lai*, Xingye Xu*, Zhen Zhang, Zhenyu Yang, Hao Yuan, and Lei Wang. "MiDSFN: A novel signature fusion learning network for potential drug-microbe associations prediction." **Code**

TL;DR | Developed MiDSFN, a feature-independent multimodal framework for cold-start drug-microbe association prediction that avoids association-matrix-derived features, reduces information leakage, and improves evaluation under cross-dataset transfer and unverified negative labels.

Projects

Project Lead, A Novel Combined Neural Network Model for Predicting Microbe-Drug Associations Based on Bilinear Attention Networks Nov. 2024 - Jun. 2025

Awards

Second Prize, National College Student Mathematics Competition (Mathematics Group B)	Dec. 2024
Second Prize, Hunan Provincial College Student Mathematics Competition (Mathematics Group B)	Dec. 2024
Honorable Mention, COMAP MCM/ICM	Feb. 2025
Second Prize, Chia Tai Cup National College Student Market Survey and Analysis Competition	Apr. 2025
Third Prize, Challenge Cup Hunan Provincial Competition (Extracurricular Academic Works)	June 2025
Third Prize, Hunan Provincial College Student Market Survey and Analysis Competition	Aug. 2025
Third Prize, Changsha College Student Data Elements-Empowered AI Innovation and Entrepreneurship Competition	Dec. 2025